SAMPLE

Famous Astronomers: Annie Jump Cannon

Classifying the Stars

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Introduction

The volume opens with a woman at a desk, a magnifying glass in one hand and a glass photographic plate in the other, working through a stack of stellar spectra at a pace no astronomer before her had ever matched. Annie Jump Cannon classified more stars than any single human being in the history of the science, roughly three hundred and fifty thousand of them, and the alphabetical sequence she built, running O through M, still orders every star chart printed today. The method deserves its introduction entry too, because the account will keep returning to it: a glass plate carrying a star's light spread into its colors, the eye trained to read the dark lines across the smear as a signature, a letter assigned, and the next plate, the whole empire of the science reduced to a repeated small judgment, the introduction's own standing definition of what classifying actually meant, hour after hour, for forty years.

The figure the account opens on deserves a plain entry: a woman at a desk in the Harvard College Observatory, the second floor brick room the era's photographs keep, the stacks of glass beside her, the ruled sheets in front of her, the pencil that outlasted a decade of mornings, the whole empire of the stars reduced to a room, a glass, and a hand, the account's own first portrait, the one the chapters ahead will keep returning to and enlarging.

Her working life ran from Dover, Delaware, where her mother taught her the constellations from an attic window, to the Harvard College Observatory, where she turned a roomful of glass plates into the largest stellar catalog ever assembled by eye. The rooms between those two poles hold the account: the Wilmington school years where the mathematics showed first, the Wellesley decade under Sarah Frances Whiting where the physics was learned, the scarlet fever years where the hearing went, the return to Wellesley as a teacher, the Radcliffe arrangement that opened the Harvard doors, and the plate room where the rest of the life was spent. The reader who wants the person before the science can take the standing summary the era's witnesses agreed on: cheerful, tireless, nearly deaf, and the fastest pair of eyes in the history of the discipline.

The world she entered deserves its own paragraph, because the account cannot be read honestly without it. The Harvard Computers worked in an era when the great universities admitted women to the lectures and withheld the degrees, when the scientific societies elected them rarely and honored them more rarely, and when the labor of measuring the sky was priced, in the ledgers, as clerical. That the largest stellar census ever attempted was built inside that world is the fact the context chapter will carry in full, and the introduction's own honest note is that the account tells both stories at once, the triumph and the terms, the catalog the reader will meet in every later chapter owing its existence to a woman the institution would not pay properly for forty years. She was nearly deaf for most of that career. She was also, by the testimony of everyone who watched her work, the fastest classifier of starlight the world has ever employed.

The Delaware entry deserves the first word: a shipbuilder's daughter on the Atlantic coastal plain, the attic window where the textbook opened, and the mother who read the sky as a household subject alongside the family's accounts, the whole career beginning as a lesson between two women at a window, the account's own standing origin story.

The chapters ahead follow the catalog's standing convention for a life in astronomy. The discovery chapter traces the Delaware childhood and the Wellesley education that formed her. The understanding and characteristics chapters describe the person and the practice, the ebullient manner and the tireless hand. The science chapter explains spectroscopy itself, why a star's light spread into a rainbow can be read as a signature of temperature. The observation chapter walks the reader into the plate room itself, and the measurement chapter follows the plates into print.

The context chapter places Cannon among the Harvard Computers, the women Pickering hired, and beside Williamina Fleming, her colleague and the subject of a companion volume. The modern chapter shows what her classes mean now, the research chapters follow the Draper Catalogue through its nine published volumes, The reader's own preparation belongs at the introduction's close: the mnemonic may already be in the reader's mouth, the letters may already be familiar from a classroom wall, and the account's standing promise being that the woman behind the alphabet will be less familiar than her work, the introduction's own honest wager, the reader arriving for the trivia and staying for the forty five years.

The pocket chapter keeps the small delights, the trivia and the mnemonic. The conclusion returns the account to the reader, and the reader who completes it will understand why a deaf woman from Delaware holds a permanent seat in the alphabetical heart of stellar astronomy.

A scope note on the word that organizes this volume, because the book is about a classifier and classification itself needs an introduction. To classify a star is to read its light's signature and assign it a place in a system, a judgment made in seconds at the desk but built on decades of agreement; the account uses the word in its full honest sense throughout, neither reducing Cannon's work to clerical sorting nor inflating it into theory, the truth being that her kind of sorting was the century's observational bedrock. The reader who finishes the volume will know what a spectral class is, how it was earned, and why one woman's judgment, repeated a few hundred thousand times, still underwrites the way the sky's stars are named.

One note on the account's sources, because this life is unusually well documented and unusually well remembered. The observatory's own records survive: the classification slips, the catalog proofs, the directors' correspondence, the colleagues' memoirs, and the subject's own diaries and letters, held in the archive of the institution she served for four decades; the account draws on the paper trail where it is firm and on the retellings where they are colored, flagging the difference. The volatile items, the anecdote's details and the era's gossip, carry the book's standing caution; the plate counts and the catalog dates, checked against the published volumes, carry its confidence, and the reader is told which kind of sentence is which.

Chapter 1: The Discovery of Annie Jump Cannon

The discovery chapter begins, as astronomer biographies in this catalog always do, with the childhood sky. Annie Jump Cannon was born on December 11, 1863, in Dover, Delaware, the eldest of three daughters of Wilson Cannon, a shipbuilder and state senator, and his second wife, Mary Jump. The mother is the discovery's true origin: Mary Jump taught her daughter the constellations from an old astronomy textbook, held the attic sessions where the two of them identified the stars visible from Delaware, and steered the girl toward the mathematics and chemistry and biology she would need for a science the era barely admitted women to.

The home itself deserves its own entry, because the account's later chapters keep drawing on it. Wilson Cannon's household was a Delaware political family, the father a state senator as well as a shipbuilder, the house full of the standing and the visits that standing brings, and the three daughters raised with an emphasis the era reserved for sons in most families: real schooling, real mathematics, and the expectation that the girls would use them. Mary Jump is the parent the record keeps, the mother's own sky literacy unusual for her generation, and the standing fact of the childhood being the attic partnership, mother and eldest daughter with one textbook between them, the account's first observatory being a Delaware attic and its first telescope a mother's finger pointed at the ceiling of a winter sky.

The schooling followed. Wilmington Conference Academy, where the young Cannon proved a promising student in mathematics. Then, in 1880, Wellesley College in Massachusetts, one of the finest academic schools for women in the country, where she studied physics and astronomy under Sarah Frances Whiting, one of the few women physicists in the United States. The Wellesley graduation deserves its own entry: 1884, the physics degree among the first the college granted, the valedictory address the class gave her the honor of delivering, and the standing note the account adds, the physics she learned there being the era's whole kit, mechanics and optics and the first electricity, the spectroscopy arriving only later at Whiting's side, the graduation being the certificate the career began on, the account's own first milestone in a file of them.

The interruption came next, and it belongs in this chapter honestly. Scarlet fever struck in the years after college, and the illness took most of her hearing. The loss made social life difficult, and Cannon, by her own account and everyone else's, immersed herself in work instead. The deafness was not a footnote to the career. It ran beside the whole of it, and the later chapters will show a scientist who turned a silent world into a proof that starlight can be read on paper without a sound.

The Wellesley decade's own texture deserves an entry before the grief: Whiting's physics course one of the few places in America where a woman could learn experimental physics, the college's observatory and its spectroscopy demonstrations, the young Cannon's notebooks from those years showing the careful hand the plate room would later employ, and the standing fact of the teacher's later career, Whiting staying at Wellesley while her student went on to Harvard, the account's own first mentorship, the Delaware attic's second teacher and the career's first professional one.

The return to astronomy came through grief and a letter. Her mother died in 1894, and life at home grew harder. Cannon wrote to her old instructor, Sarah Frances Whiting, asking whether a position existed. Whiting hired her as a junior physics teacher at Wellesley, let her take graduate courses in physics and astronomy, and introduced her to spectroscopy, the study of light spread into its colors. To reach a better telescope, Cannon enrolled at Radcliffe

College in 1894 as a special student, the arrangement by which Harvard professors repeated their lectures to Radcliffe women.

The Radcliffe years carry their own texture: the special student status meaning no degree at the end, the lectures attended across the divide between the women's college and the men's university, the physics and astronomy courses repeated for the Radcliffe women in the afternoons, and the standing arrangement's quiet honesty, the era educating women with one hand and withholding its own credentials with the other.

The arrangement gave her the thing she needed most: access to the Harvard College Observatory. In 1896 Edward Charles Pickering, the Observatory's director, hired her as his assistant. She joined the group now remembered as the Harvard Computers, the women hired to reduce and measure and catalog the Observatory's photographic plates. The discovery chapter's own standing note belongs here: the catalog's biographies open with a discovery because a scientific life has one, the moment the preparation meets the opportunity, and Cannon's came in 1896, a thirty two year old deaf woman with a Wellesley education and a Radcliffe arrangement, hired into the room where the Henry Draper project's plates were waiting for an eye like hers, the discovery being mutual, the account's own standing convention, the person found by the work and the work by the person.

The Wellesley years belong in the discovery chapter at full length, because the education was the life's first door. The college's founder had built a science curriculum unusual for the era, the physics and astronomy taught by a woman professor with a laboratory and a small observatory; the student from Delaware arrived in the early eighties and took the physics course, learned the spectroscopy of her era at first hand, and sat under the roof of an institution that assumed women could do exact science; the account's honest note is the doors' arithmetic, the whole of the later career traceable to a college that said yes when the century mostly said no, the discovery chapter keeping the door's frame as carefully as the life that walked through it.

The home decade belongs in the discovery chapter too, because the road between the diploma and the observatory ran through a Delaware house and a long interruption. The years after college held family duty and illness, the scarlet fever that took most of her hearing, the quiet decade of photographs and travel and the local society's lectures; the return ran through her old college's advanced course in the mid nineties, the professor of physics encouraging the student of forty toward the new profession, and the transfer to the university at Cambridge following within the year. The chapter's honest judgment: the decade reads like a detour and worked like a ferment, the classifier's patience and the photographer's eye both grown in the years the career looked stalled.